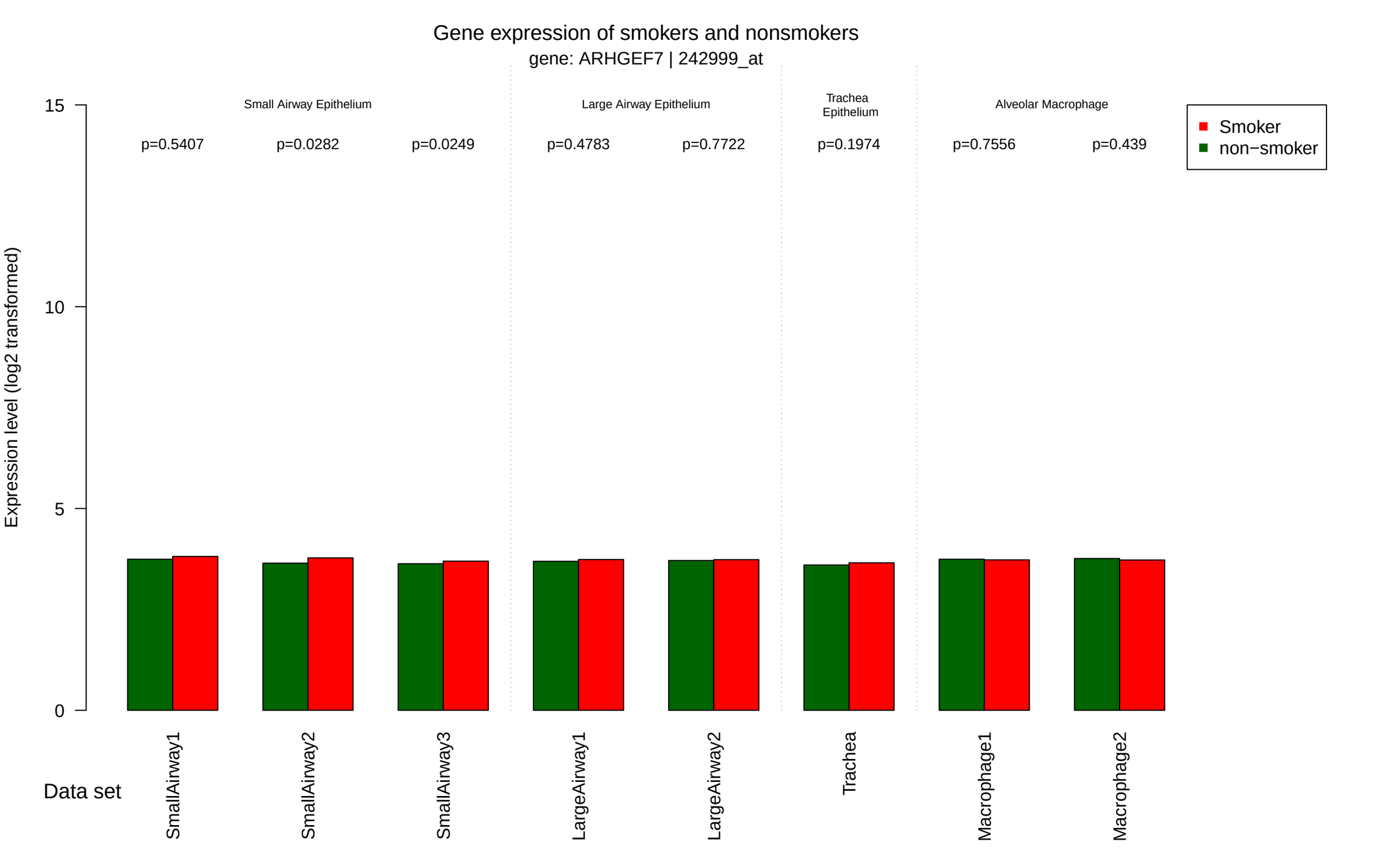
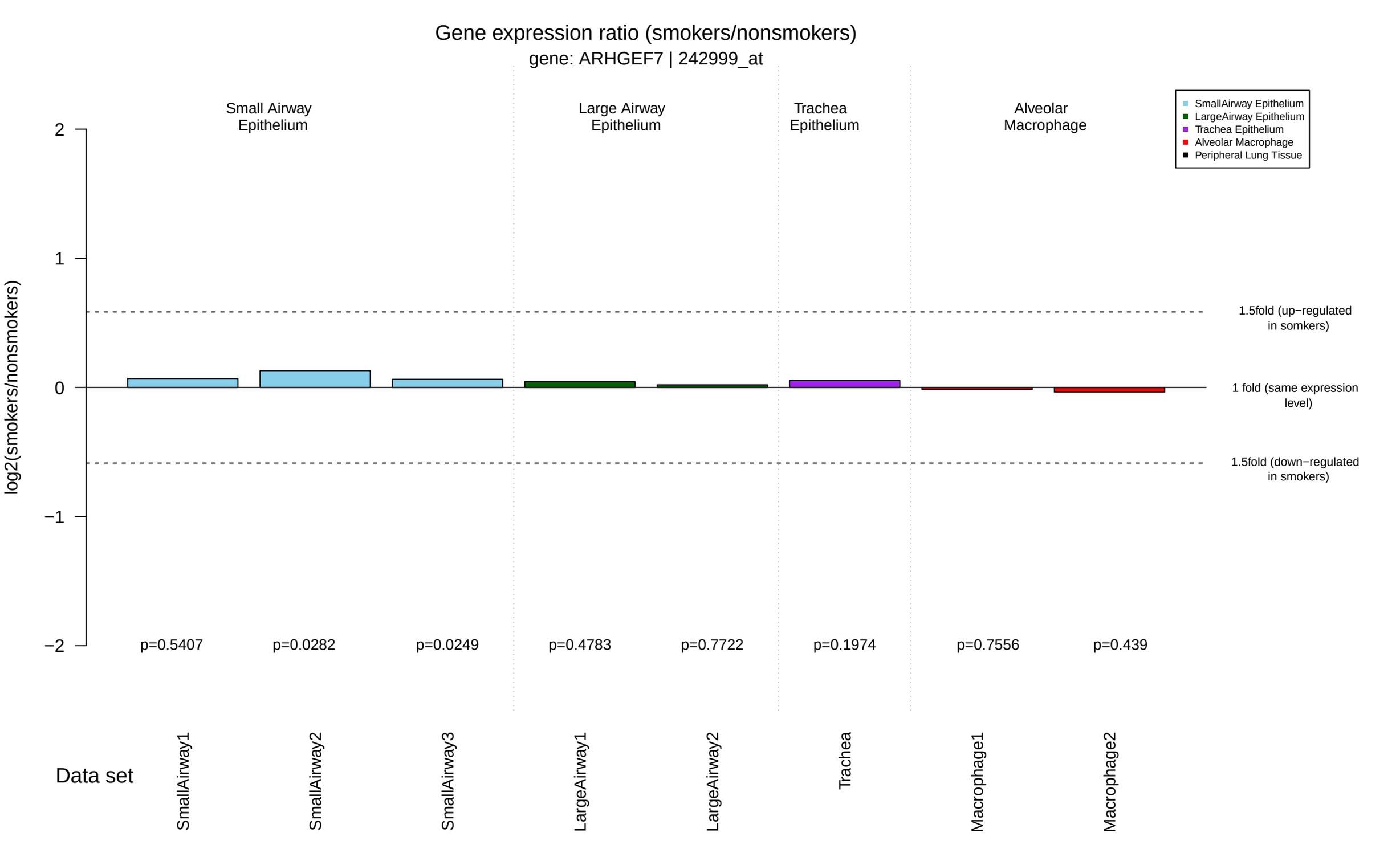
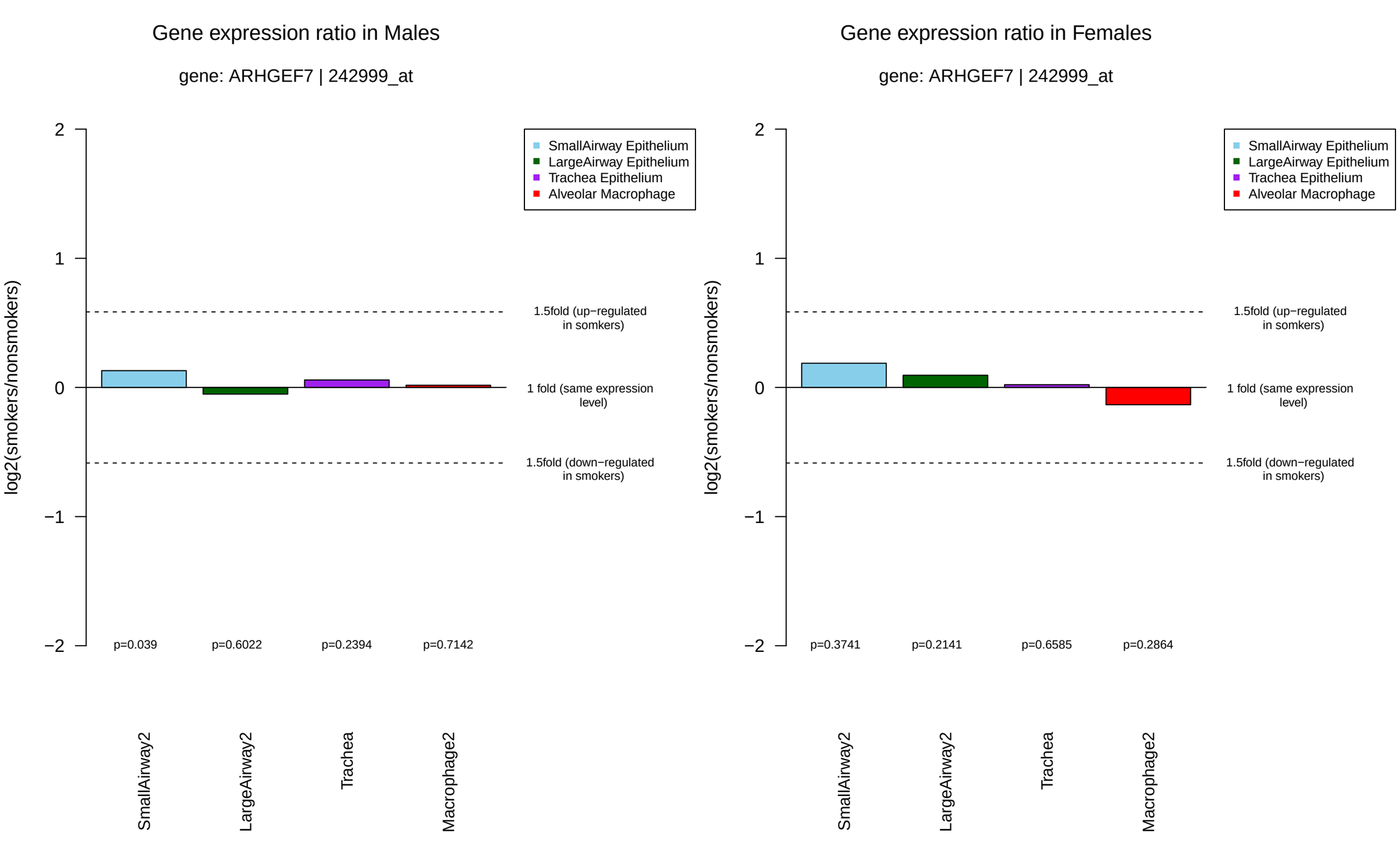
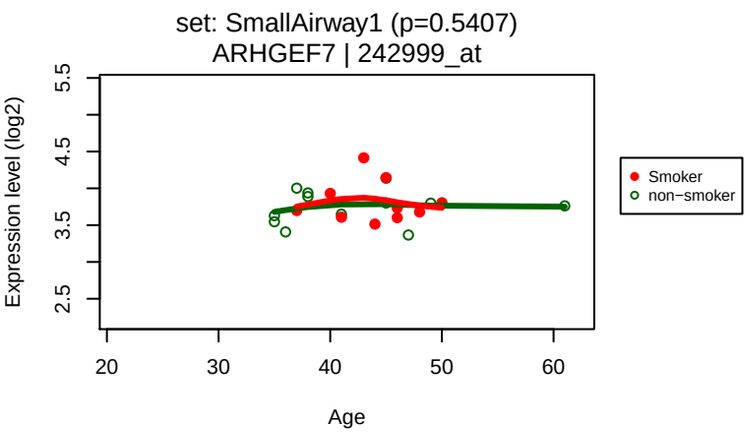
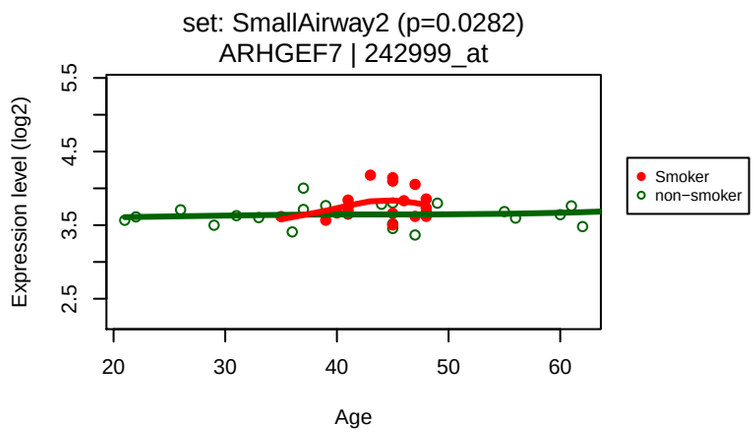
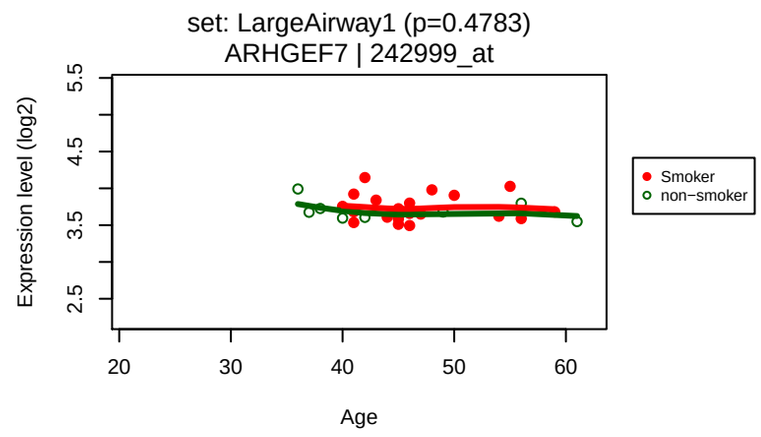

Gene expression of smokers and nonsmokers
gene: ARHGEF7 | 242999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5407
p=0.0282
p=0.0249
p=0.4783
p=0.7722
p=0.1974
p=0.7556
p=0.439
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

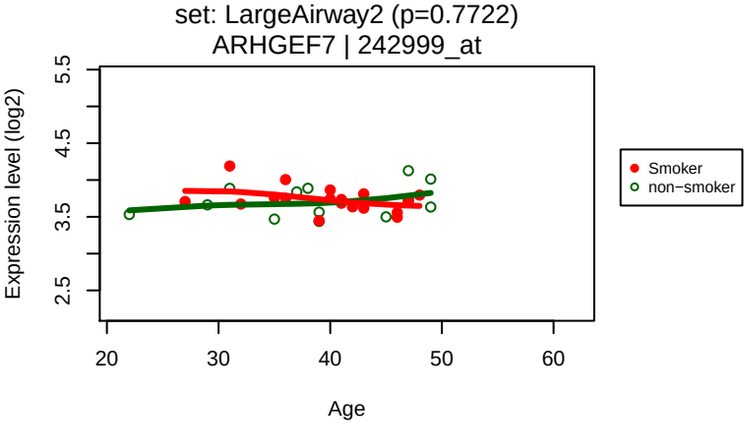

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF7 | 242999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5407
p=0.0282
p=0.0249
p=0.4783
p=0.7722
p=0.1974
p=0.7556
p=0.439
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

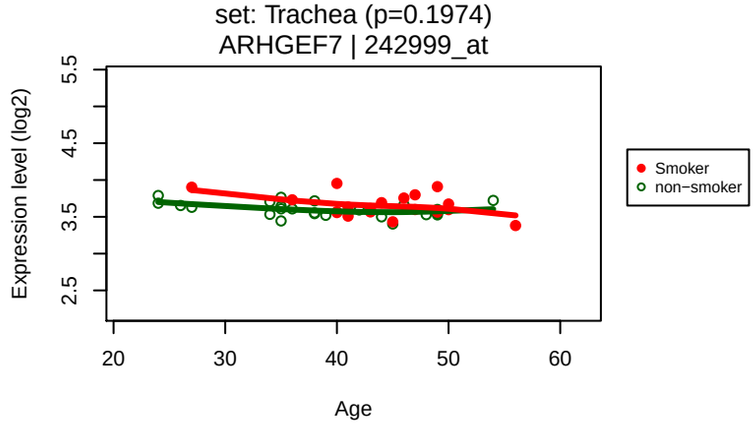

Gene expression ratio in Males
gene: ARHGEF7 | 242999_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.039
p=0.6022
p=0.2394
p=0.7142
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF7 | 242999_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3741
p=0.2141
p=0.6585
p=0.2864
SmallAirway2
LargeAirway2
Trachea
Macrophage2

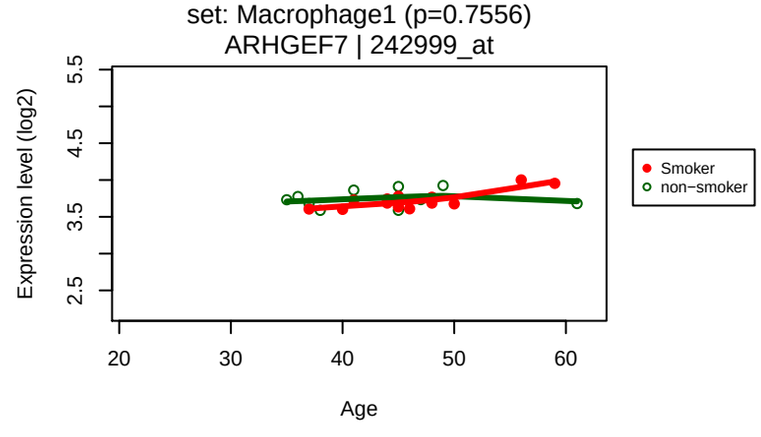

set: SmallAirway1 (p=0.5407)
ARHGEF7 | 242999_at
Expression level (log2)
Age
Smoker
non−smoker

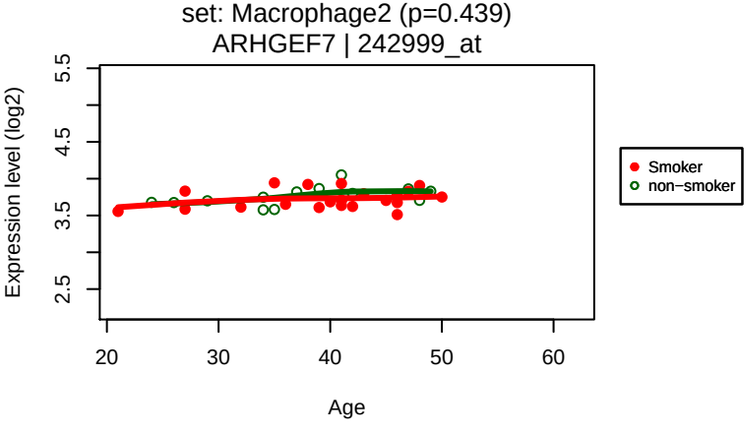

set: SmallAirway2 (p=0.0282)
ARHGEF7 | 242999_at
Expression level (log2)
Age
Smoker
non−smoker

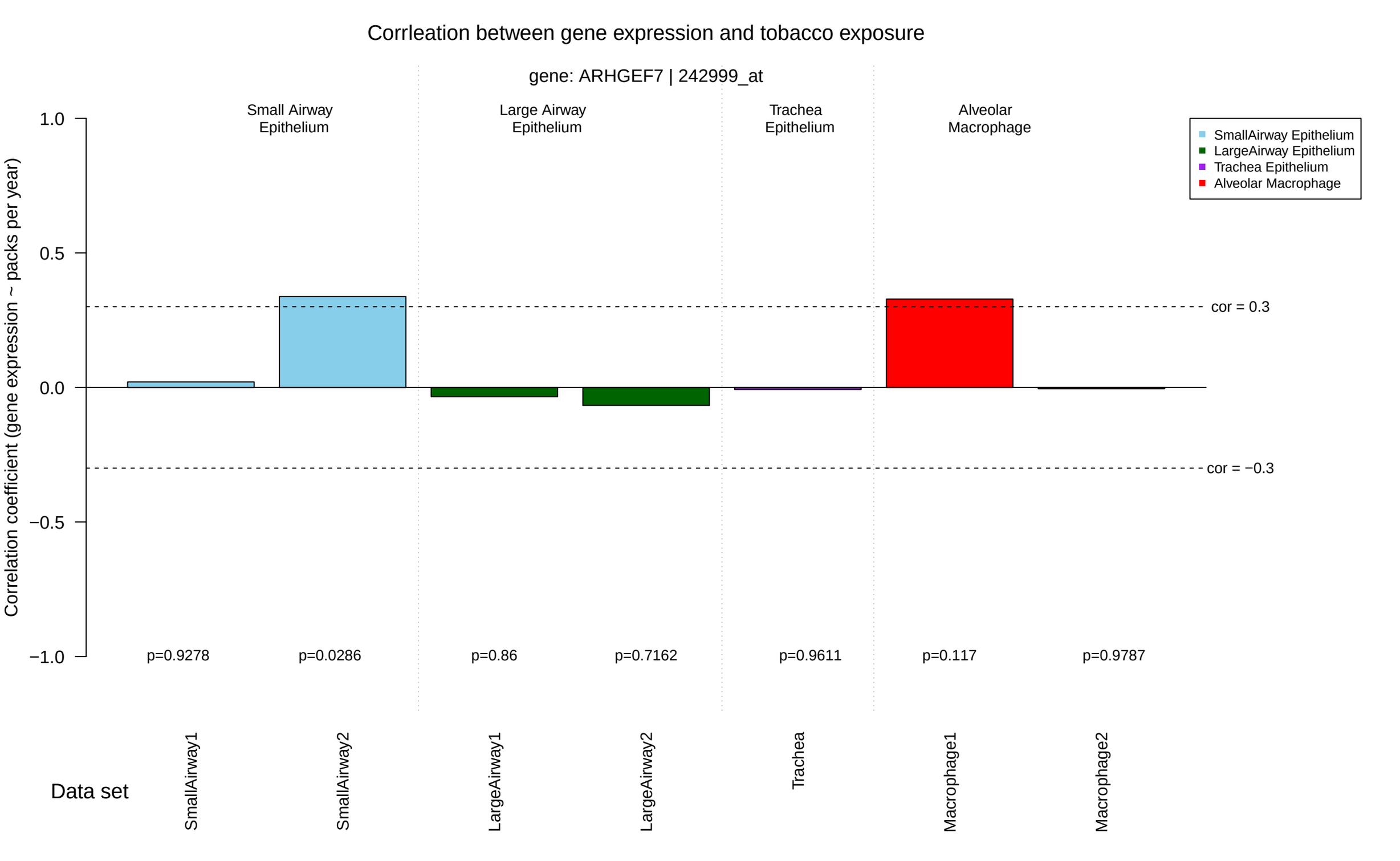

set: LargeAirway1 (p=0.4783)
ARHGEF7 | 242999_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7722)
ARHGEF7 | 242999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1974)
ARHGEF7 | 242999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7556)
ARHGEF7 | 242999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.439)
ARHGEF7 | 242999_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF7 | 242999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9278
p=0.0286
p=0.86
p=0.7162
p=0.9611
p=0.117
p=0.9787
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2